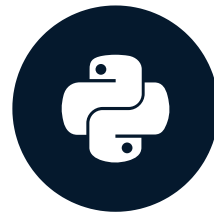


Introduction to Hugging Face

WORKING WITH HUGGING FACE



Jacob H. Marquez
Lead Data Engineer

What is Hugging Face?



- Collaboration platform for ML and AI
- Central hub with 1M+ models and datasets
- Supports text, vision, and audio tasks
- Makes advanced AI accessible to all

¹ <https://huggingface.co/>

Course overview

- Designed for Developers, Data Scientists, and ML Practitioners
- Master Hugging Face Hub, models, and datasets
- Build real-world AI pipelines



Large Language Models

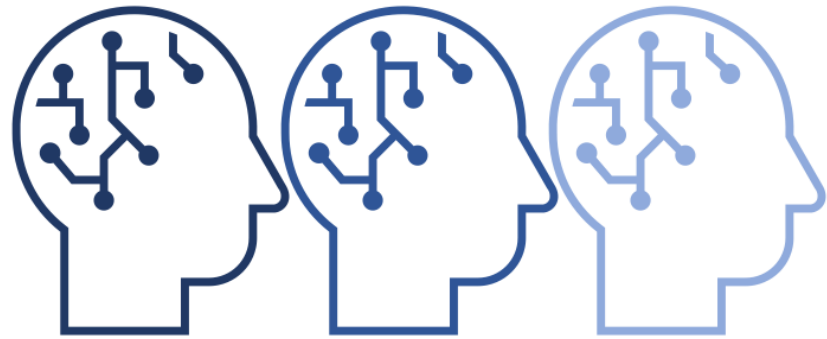
- Powerful tools for human-like text generation
- Trained on massive datasets
- Popular options are GPT and Llama



ChatGPT

LLaMA
by  **Meta**

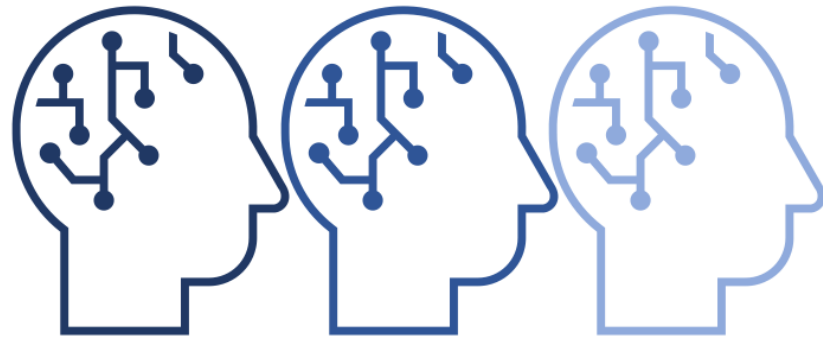
Benefits of Hugging Face



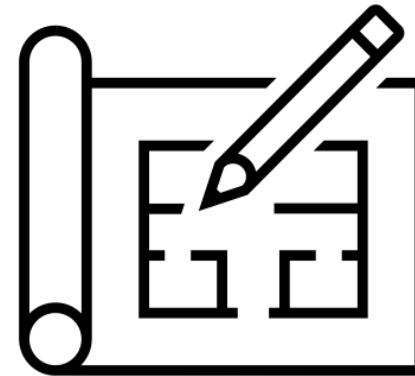
Access to Models

- Faster workflows

Benefits of Hugging Face



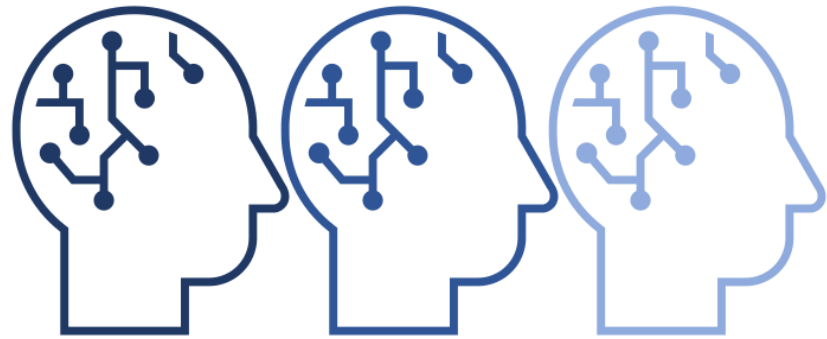
Access to Models



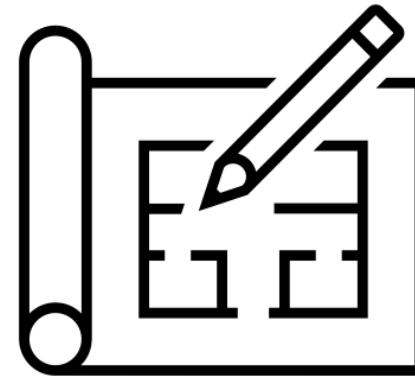
Frameworks

- Faster workflows
- Complete ecosystem to simplify workflows

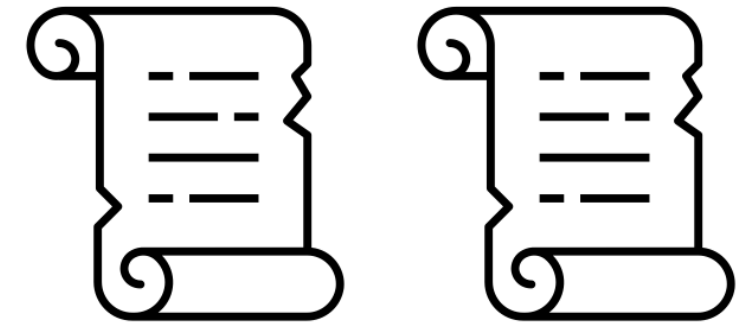
Benefits of Hugging Face



Access to Models



Frameworks

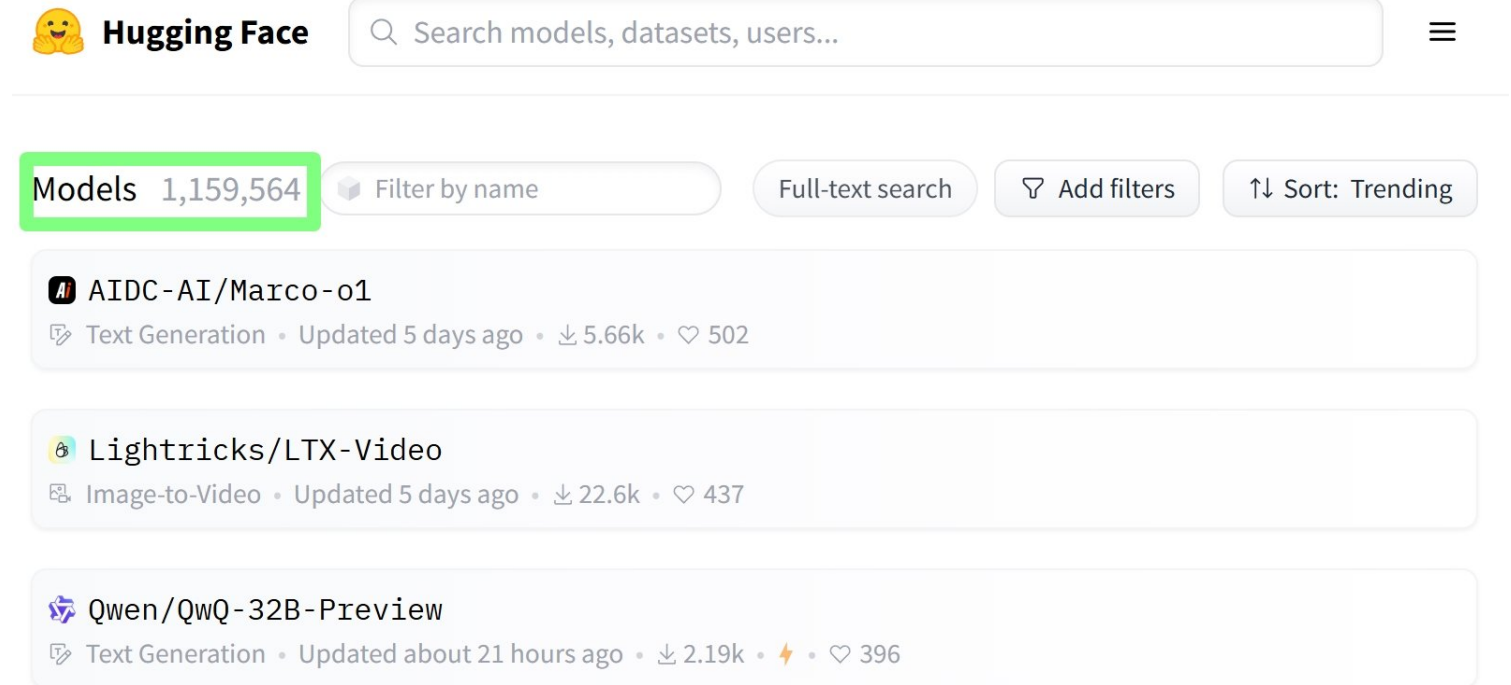


Documentation

- Faster workflows
- Complete ecosystem to simplify workflows
- Community support

Discovering Models on Hugging Face Hub

- Explore models via website



- Access models using the API



Exploring Models Programmatically

```
pip install huggingface_hub
```

```
from huggingface_hub import HfApi

api = HfApi()
models = list(api.list_models(limit=3))
print(models)
```

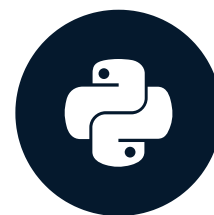
```
[ModelInfo(id='Datou1111/shou_xin', author=None, ...),
 ModelInfo(id='Qwen/QwQ-32B-Preview', author=None, ...),
 ModelInfo(id='OpenGVLab/InternVL2_5-78B', author=None, ...)]
```

¹ https://github.com/huggingface/huggingface_hub

Let's practice!
WORKING WITH HUGGING FACE

Working with Models from Hugging Face Hub

WORKING WITH HUGGING FACE



Jacob H. Marquez
Lead Data Engineer

Introduction to the Transformers Library

- Simplifies working with pre-trained models



Transformers

build failing license Apache-2.0 website online release v4.46.3
Contributor Covenant v2.0 adopted DOI 10.5281/zenodo.7391177

English | 简体中文 | 繁體中文 | 한국어 | Español | 日本語 | हिन्दी | Русский |
Português | తెలుగు | Français | Deutsch | Tiếng Việt | اردو | العربية |

State-of-the-art Machine Learning for JAX, PyTorch and TensorFlow



**Part of the
Hugging Face course!**

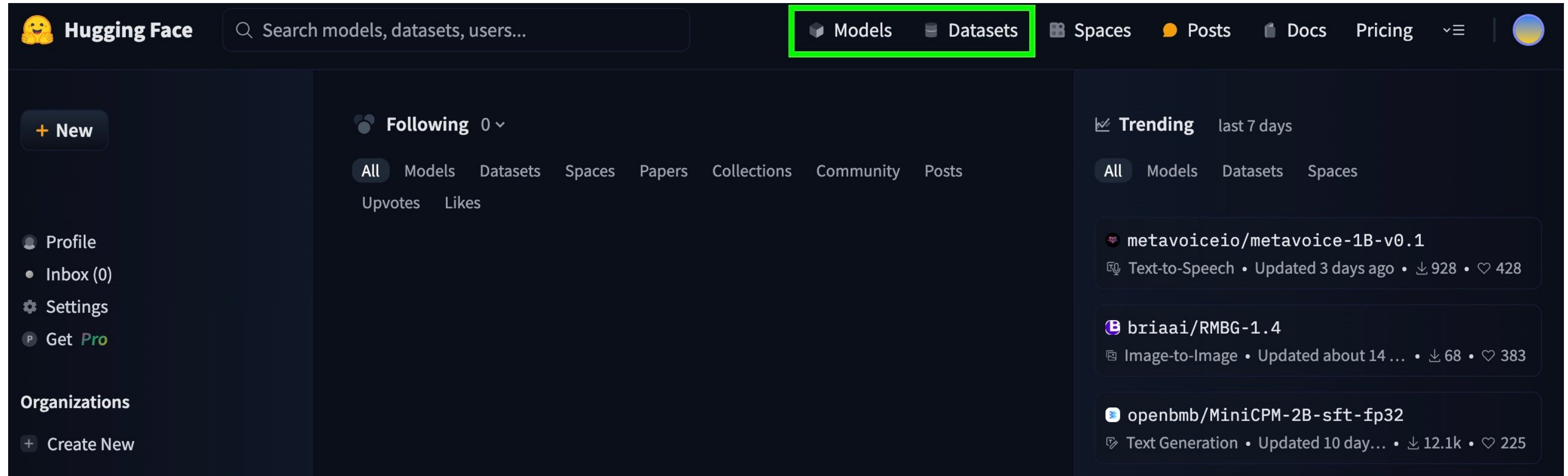
¹ <https://github.com/huggingface/transformers>

How Pre-Trained Models Are Used

- Analyze text, images, and audio
- Summarize documents and recognize speech
- Enable autonomous driving by detecting objects

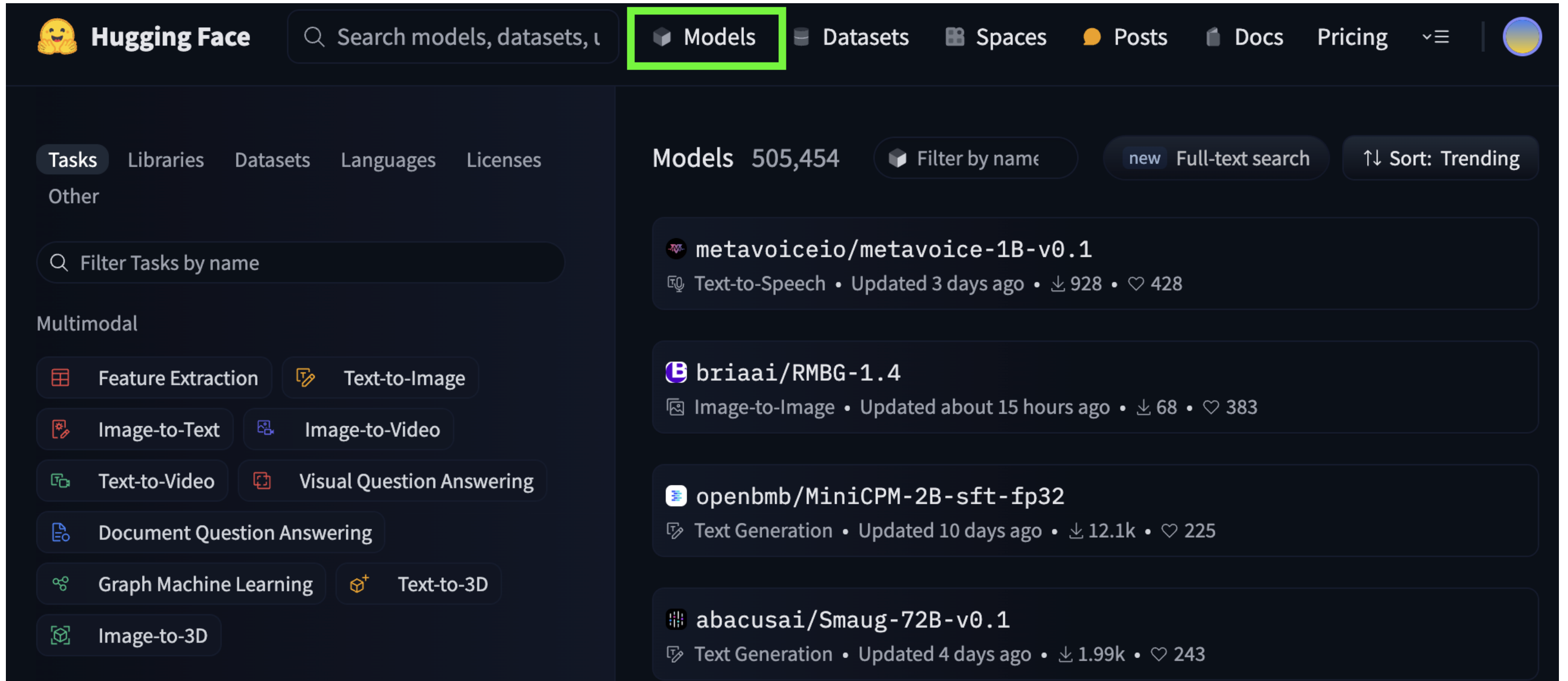


Navigating the Hub



¹ <https://huggingface.co/>

Searching for models

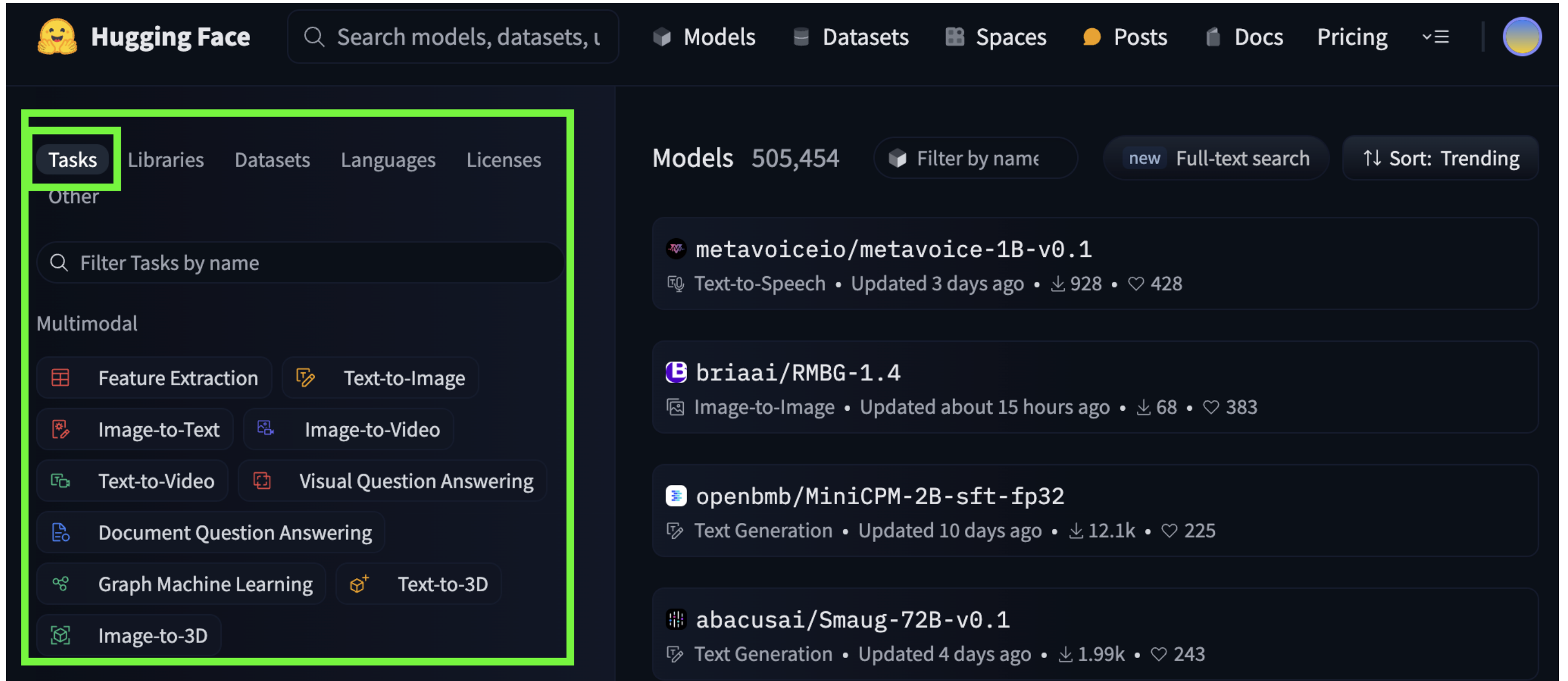


The screenshot shows the Hugging Face website interface. At the top, the 'Hugging Face' logo is on the left, followed by a search bar containing 'Search models, datasets, l'. To the right of the search bar, the 'Models' tab is highlighted with a green border. Other navigation tabs include 'Datasets', 'Spaces', 'Posts', 'Docs', and 'Pricing'. Below the navigation bar, the left sidebar contains a list of categories: 'Tasks', 'Libraries', 'Datasets', 'Languages', 'Licenses', and 'Other'. Under 'Tasks', there is a search bar 'Filter Tasks by name' and a section titled 'Multimodal' with buttons for 'Feature Extraction', 'Text-to-Image', 'Image-to-Text', 'Image-to-Video', 'Text-to-Video', 'Visual Question Answering', 'Document Question Answering', 'Graph Machine Learning', and 'Text-to-3D', and 'Image-to-3D'. The main content area on the right is titled 'Models 505,454' and includes filters like 'Filter by name', 'new', 'Full-text search', and 'Sort: Trending'. It displays a list of model cards, each with a repository name, task type, update time, and download/like counts.

Model Name	Task	Updated	Downloads	Likes
metavoiceio/metavoice-1B-v0.1	Text-to-Speech	Updated 3 days ago	928	428
briaai/RMBG-1.4	Image-to-Image	Updated about 15 hours ago	68	383
openbmb/MiniCPM-2B-sft-fp32	Text Generation	Updated 10 days ago	12.1k	225
abacusai/Smaug-72B-v0.1	Text Generation	Updated 4 days ago	1.99k	243

¹ <https://huggingface.co/models>

Searching for models



The screenshot shows the Hugging Face website interface. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Posts, Docs, and Pricing. The left sidebar is highlighted with a green border and contains the following sections:

- Tasks** (highlighted with a green box)
- Libraries
- Datasets
- Languages
- Licenses
- Other

Under the 'Tasks' section, there is a search bar labeled 'Filter Tasks by name' and a list of tasks under the 'Multimodal' category:

- Feature Extraction
- Text-to-Image
- Image-to-Text
- Image-to-Video
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning
- Text-to-3D
- Image-to-3D

The main content area displays a list of models under the 'Models' tab, which shows 505,454 models. The list includes:

- metavoiceio/metavoice-1B-v0.1**
Text-to-Speech • Updated 3 days ago • 928 downloads • 428 likes
- briaai/RMBG-1.4**
Image-to-Image • Updated about 15 hours ago • 68 downloads • 383 likes
- openbmb/MiniCPM-2B-sft-fp32**
Text Generation • Updated 10 days ago • 12.1k downloads • 225 likes
- abacusai/Smaug-72B-v0.1**
Text Generation • Updated 4 days ago • 1.99k downloads • 243 likes

¹ <https://huggingface.co/models>

Searching for models

The screenshot shows the Hugging Face website interface. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Posts, Docs, and Pricing. The left sidebar contains a list of categories: Tasks, Libraries, Datasets, Languages (highlighted with a green box), Licenses, and Other. Below this is a search bar for filtering languages by name. A grid of language buttons is displayed, including English, Chinese, French, German, Spanish, Japanese, Korean, Russian, Italian, Portuguese, Arabic, Hindi, Turkish, Dutch, Swedish, multilingual, Polish, Indonesian, Vietnamese, Finnish, Enawené-Nawé, Romanian, Thai, Ukrainian, and Persian. The main content area shows a list of models with the following details:

- Models** 505,454
- Filter by name**
- new** Full-text search
- Sort: Trending**
- metavoiceio/metavoice-1B-v0.1**
Text-to-Speech • Updated 3 days ago • 928 • 428
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Text Generation • Updated 4 days ago • 1.99k • 243

¹ <https://huggingface.co/models>

Searching for models

The screenshot shows the Hugging Face website interface. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Posts, Docs, and Pricing. The left sidebar contains a list of categories: Tasks, Libraries (highlighted with a green box), Datasets, Languages, Licenses, and Other. Below the sidebar, there is a search bar for filtering libraries by name and a grid of library tags such as PyTorch, TensorFlow, JAX, Transformers, TensorBoard, Safetensors, Diffusers, PEFT, Stable-Baselines3, ONNX, Unity ML-Agents, GGUF, Sentence Transformers, Keras, Timm, Flair, Sample Factory, SetFit, Adapters, Transformers.js, spaCy, ESPnet, fastai, Core ML, and NeMo. The main content area displays a list of models with their names, categories, update times, and download/like counts. The models listed are: metavoiceio/metavoice-1B-v0.1 (Text-to-Speech, Updated 3 days ago, 928 downloads, 428 likes), briaai/RMBG-1.4 (Image-to-Image, Updated about 15 hours ago, 68 downloads, 383 likes), openbmb/MiniCPM-2B-sft-fp32 (Text Generation, Updated 10 days ago, 12.1k downloads, 225 likes), and abacusai/Smaug-72B-v0.1 (Text Generation, Updated 4 days ago, 1.99k downloads, 243 likes).

¹ <https://huggingface.co/models>

Model cards

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Posts](#) [Docs](#) [Pricing](#) ⌵

 **openai/whisper-large-v3** 📄 👍 like 1.63k

[👤 Automatic Speech Recognition](#) [🧠 Transformers](#) [🔥 PyTorch](#) [🦋 JAX](#) [🛡️ Safetensors](#) [🌐 99 languages](#) [whisper](#) [audio](#)

[hf-asr-leaderboard](#) [🎤 Inference Endpoints](#) [📄 arxiv:2212.04356](#) [📄 arxiv:2311.00430](#) [🏠 License: apache-2.0](#)

[📄 Model card](#) [📁 Files](#) [👏 Community 80](#)

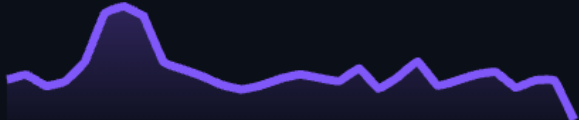
⋮ [🔗 Train](#) [🚀 Deploy](#) [📄 Use in Transformers](#)

[✎ Edit model card](#)

Whisper

Whisper is a pre-trained model for automatic speech recognition (ASR) and speech translation. Trained on 680k hours of labelled data, Whisper models demonstrate a strong ability to generalise to

Downloads last month
931,146



[🛡️ Safetensors ⓘ](#) [Model size](#) **1.54B params**

[Tensor type](#) **FP16** [↗](#)

¹ <https://huggingface.co/openai/whisper-large-v3>

Running a Basic Pipeline

```
from transformers import pipeline

my_pipeline = pipeline(
    "text-classification",
    model="distilbert-base-uncased-finetuned-sst-2-english"
)

print(my_pipeline("DataCamp is awesome!"))
```

```
[{'label': 'POSITIVE', 'score': 0.99}]
```

Adjusting Pipeline Parameters

```
from transformers import pipeline

my_pipeline = pipeline("text-generation", model="gpt2")

results = my_pipeline("What if AI", max_length=10, num_return_sequences=2)

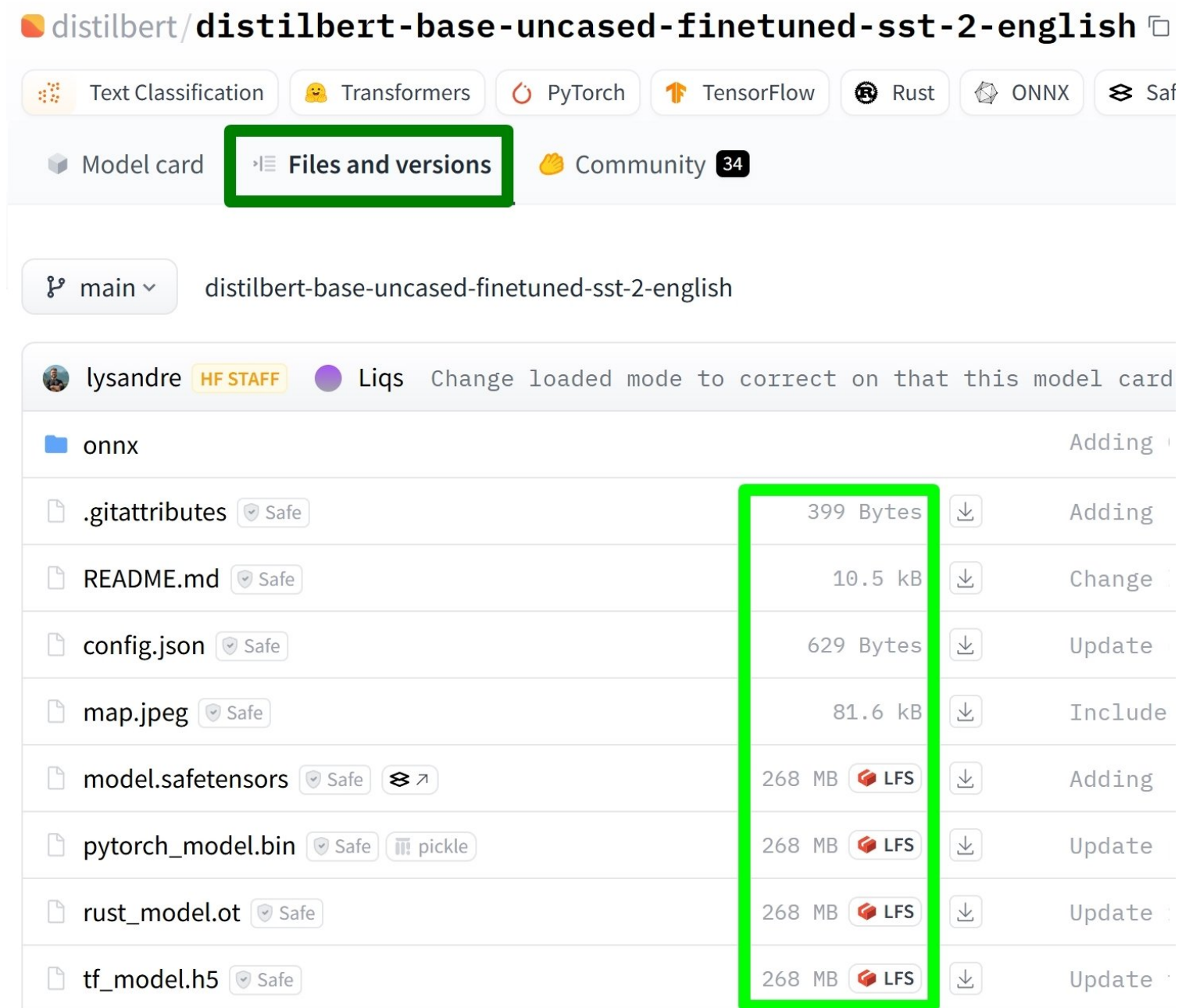
for result in results:
    print(result['generated_text'])
```

What if AI had never existed?

What if AI could be really smarter than us?

When and why to save models

- Save models locally for:
 - Offline access
 - Custom modifications
 - Large-scale deployments
- ☐ Check model size on the Hub before saving!



The screenshot shows the Hugging Face model card for `distilbert-base-uncased-finetuned-sst-2-english`. The **Files and versions** tab is selected and highlighted with a green box. Below the tab, a table lists the files in the repository. A green box highlights the file sizes column, showing that the model files are 268 MB in size.

File	Size	Format	Safe	Action
onnx	-	-	-	Adding
.gitattributes	399 Bytes	-	Safe	Adding
README.md	10.5 kB	-	Safe	Change
config.json	629 Bytes	-	Safe	Update
map.jpeg	81.6 kB	-	Safe	Include
model.safetensors	268 MB	LFS	Safe	Adding
pytorch_model.bin	268 MB	LFS	Safe	Update
rust_model.ot	268 MB	LFS	Safe	Update
tf_model.h5	268 MB	LFS	Safe	Update

¹ <https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english/tree/main>

Saving a model locally

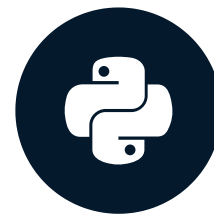
```
my_pipeline = pipeline("text-classification",  
                        model="distilbert-base-uncased-finetuned-sst-2-english")  
  
my_pipeline.save_pretrained("/path/to/saved_model_directory")  
  
reloaded_pipeline = pipeline("text-classification",  
                             model="/path/to/saved_model_directory")  
  
print(reloaded_pipeline("Hugging Face is great!"))
```

```
[{'label': 'POSITIVE', 'score': 0.9998798370361328}]
```


Let's practice!
WORKING WITH HUGGING FACE

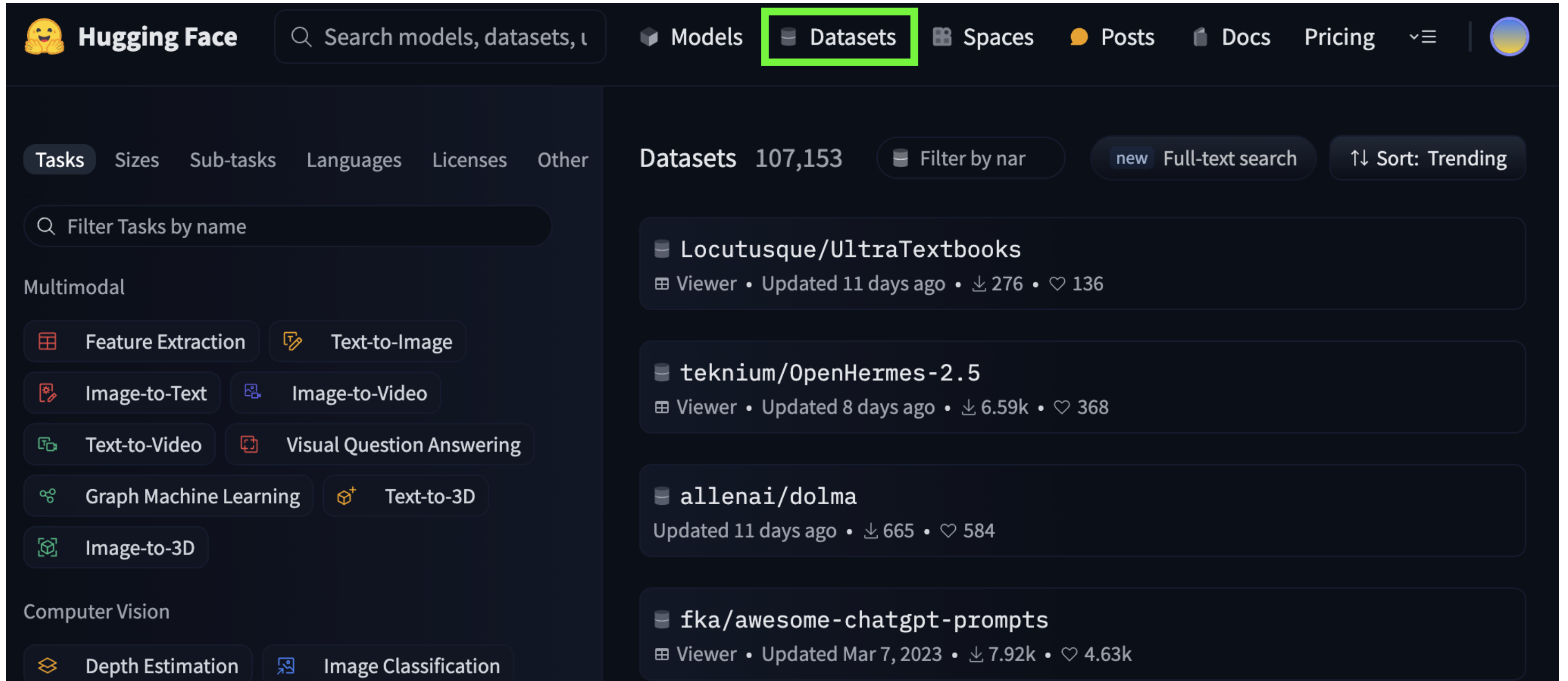
Hugging Face Datasets

WORKING WITH HUGGING FACE



Jacob H. Marquez
Lead Data Engineer

Datasets in Hugging Face

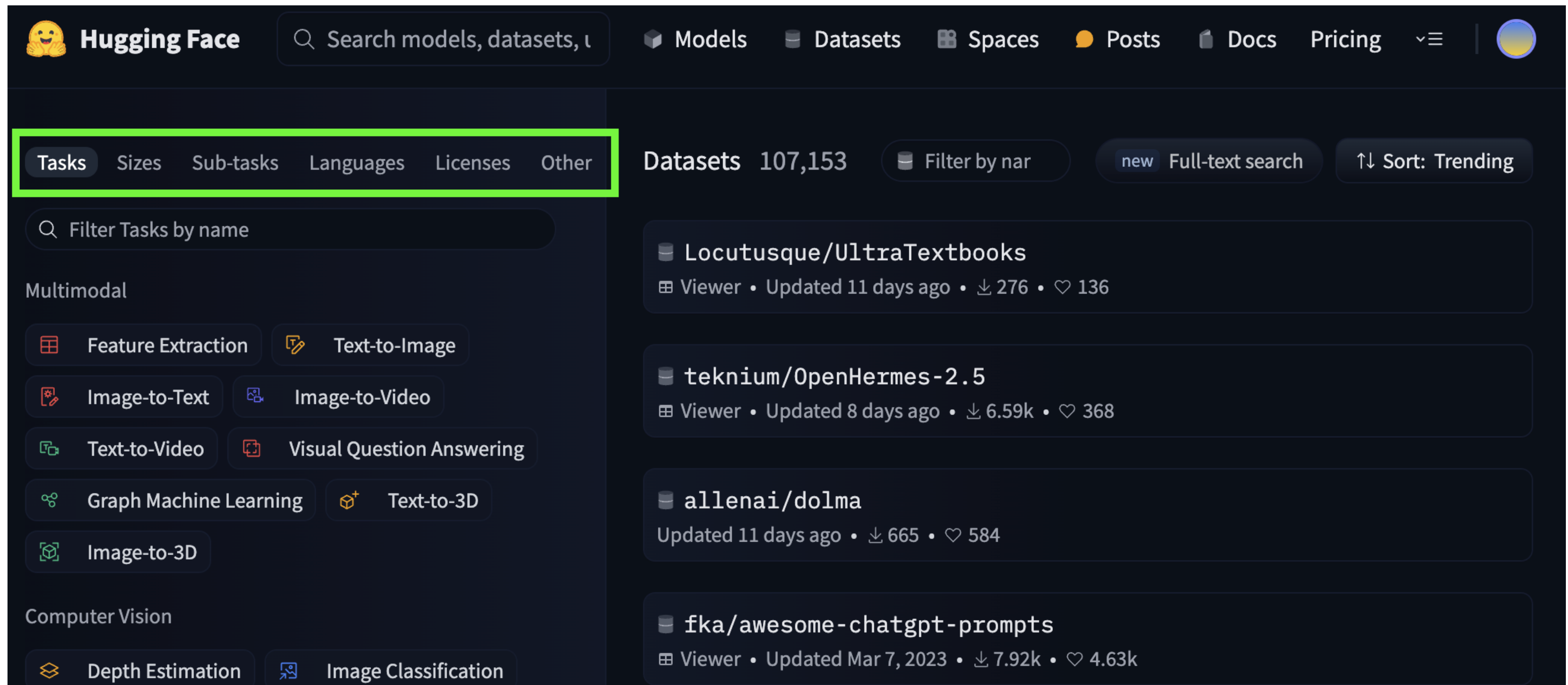


The screenshot shows the Hugging Face website's 'Datasets' page. The top navigation bar includes the Hugging Face logo, a search bar, and links to Models, Datasets (highlighted with a green box), Spaces, Posts, Docs, and Pricing. The left sidebar contains filters for Tasks, Sizes, Sub-tasks, Languages, Licenses, and Other, along with a search bar for tasks and a list of task categories like Multimodal, Feature Extraction, Text-to-Image, Image-to-Text, Image-to-Video, Text-to-Video, Visual Question Answering, Graph Machine Learning, Text-to-3D, Image-to-3D, Computer Vision, Depth Estimation, and Image Classification. The main content area displays a list of datasets with the following details:

- Datasets 107,153** (Filter by nar, new Full-text search, Sort: Trending)
- Locutusque/UltraTextbooks**
 - Viewer • Updated 11 days ago • 276 • 136
- teknium/OpenHermes-2.5**
 - Viewer • Updated 8 days ago • 6.59k • 368
- allenai/dolma**
 - Updated 11 days ago • 665 • 584
- fka/awesome-chatgpt-prompts**
 - Viewer • Updated Mar 7, 2023 • 7.92k • 4.63k

¹ <https://huggingface.co/datasets>

Searching for datasets



The screenshot shows the Hugging Face website's 'Datasets' page. The 'Tasks' filter is highlighted with a red box. The page displays a list of datasets, including 'Locutusque/UltraTextbooks', 'teknium/OpenHermes-2.5', 'allenai/dolma', and 'fka/awesome-chatgpt-prompts'. The 'Tasks' filter is currently selected, showing a list of tasks such as Feature Extraction, Text-to-Image, Image-to-Text, Image-to-Video, Text-to-Video, Visual Question Answering, Graph Machine Learning, Text-to-3D, Image-to-3D, Depth Estimation, and Image Classification.

Hugging Face Search models, datasets, ... Models Datasets Spaces Posts Docs Pricing

Tasks Sizes Sub-tasks Languages Licenses Other

Filter Tasks by name

Multimodal

- Feature Extraction
- Text-to-Image
- Image-to-Text
- Image-to-Video
- Text-to-Video
- Visual Question Answering
- Graph Machine Learning
- Text-to-3D
- Image-to-3D

Computer Vision

- Depth Estimation
- Image Classification

Datasets 107,153 Filter by nar new Full-text search Sort: Trending

- Locutusque/UltraTextbooks**
Viewer • Updated 11 days ago • 276 • 136
- teknium/OpenHermes-2.5**
Viewer • Updated 8 days ago • 6.59k • 368
- allenai/dolma**
Updated 11 days ago • 665 • 584
- fka/awesome-chatgpt-prompts**
Viewer • Updated Mar 7, 2023 • 7.92k • 4.63k

¹ <https://huggingface.co/datasets>

Dataset cards

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Posts](#) [Docs](#) [Pricing](#) 



 **Datasets: imdb** 

 like 160

Tasks:  Text Classification

Sub-tasks: sentiment-classification

Languages:  English

Multilinguality: monolingual

Size Categories: 10K<n<100K

Language Creators: expert-generated

Annotations Creators: expert-generated

Source Datasets: original

Tags:  Croissant

License:  other

 Dataset card

 Viewer

 Files

 Community 6

 Dataset Viewer

 Auto-converted to Parquet

 API

 View in Dataset Viewer

Split

train · 25k rows



Downloads last month 273,080

 Use in Datasets library

 Edit dataset card

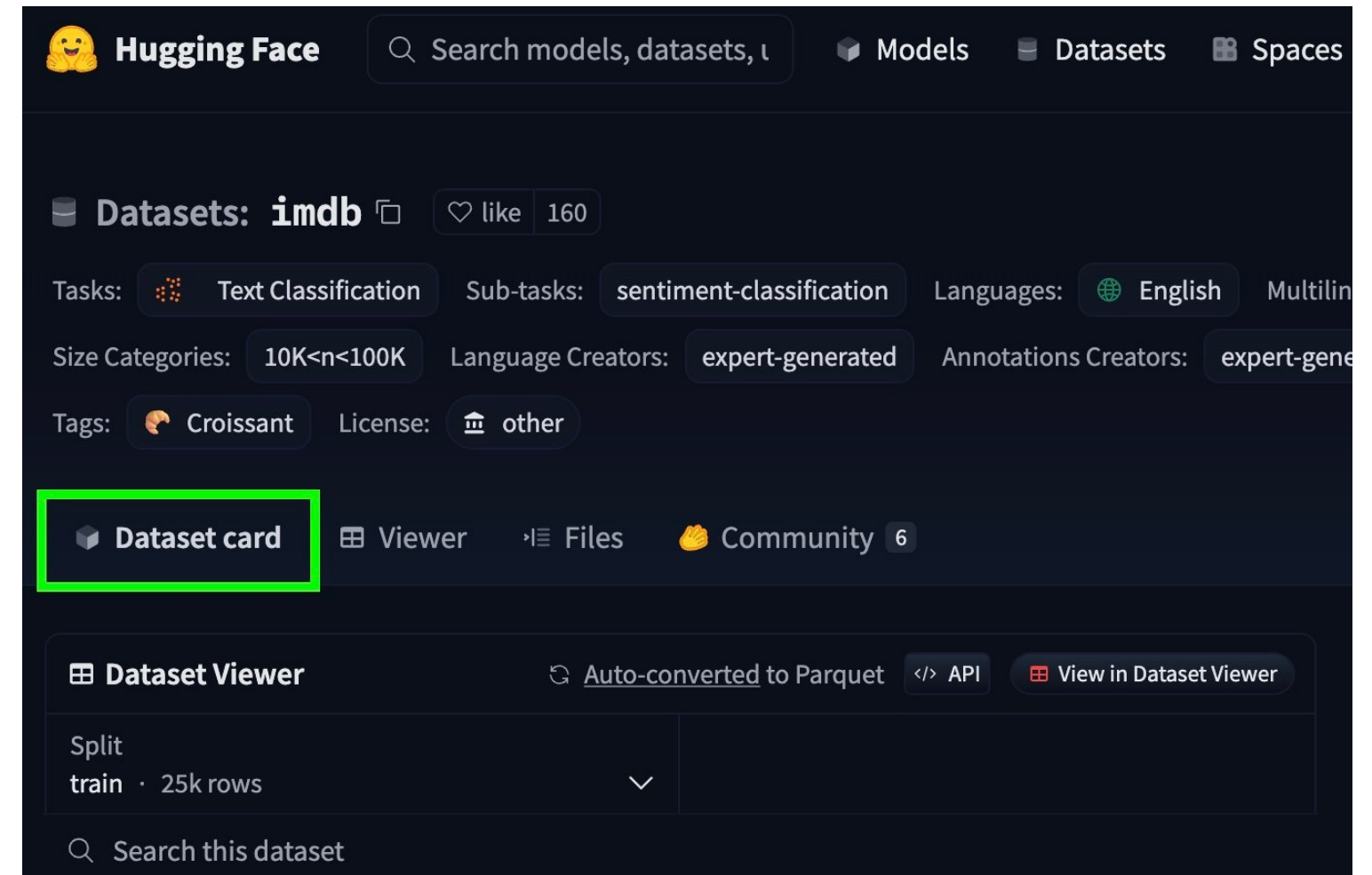
¹ <https://huggingface.co/datasets/imdb>



WORKING WITH HUGGING FACE

Dataset cards

- Path
- Description
- Dataset structure
- An example
- Field metadata



¹ <https://huggingface.co/datasets/imdb>

Dataset cards

Datasets: **imdb**

like

160

Dataset card

Viewer

Files

Community 6

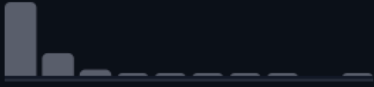
Split

train · 25k rows

Search this dataset

text

string · *lengths*



5213.7k

label

class label



2 classes

I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that surrounded it when it was first released in 1967. I also heard that at first it was seized by U.S. customs if it ever tried to enter this country,...	0 neg
"I Am Curious: Yellow" is a risible and pretentious steaming pile. It doesn't matter what one's political views are because this film can hardly be taken seriously on any level. As for the claim that frontal male nudity is an...	0 neg
If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story. One might feel virtuous for sitting thru it because it touches on so many IMPORTANT issues but it...	0 neg
This film was probably inspired by Godard's Masculin, féminin and I urge you to see that film instead. The film has two strong elements and those are, (1) the realistic acting (2) the impressive, undeservedly good, photo...	0 neg

¹ <https://huggingface.co/datasets/imdb>

 datacamp

WORKING WITH HUGGING FACE

Installing Datasets Package

```
pip install datasets
```

- ☐ Access
- ☐ Download
- ☒ Mutate
- ☐ Use
- ☐ Share



¹ <https://huggingface.co/docs/datasets/index>

Inspecting a dataset

```
from datasets import load_dataset_builder

# Load dataset metadata
data_builder = load_dataset_builder("imdb")

# Access dataset size
dataset_size_mb = data_builder.info.dataset_size / (1024 ** 2)

print(f"Dataset size: {round(dataset_size_mb, 2)} MB")
```

```
Dataset size: 127.02 MB
```

¹ https://huggingface.co/docs/datasets/load_hub

Downloading a dataset

```
from datasets import load_dataset  
  
data = load_dataset("imdb")
```

Split parameter

```
data = load_dataset("imdb", split="train")
```

¹ <https://huggingface.co/docs/datasets/v2.15.0/loading>

Apache Arrow dataset formats

Row-based			Column-based		
Row 1		1331246660	session_id		1331246660
		3/8/2012 2:44PM			1331246351
		99.155.155.225			1331244570
Row 2		1331246351			1331261196
		3/8/2012 2:38PM	timestamp		3/8/2012 2:44PM
		65.87.165.114			3/8/2012 2:38PM
Row 3		1331244570			3/8/2012 2:09PM
		3/8/2012 2:09PM			3/8/2012 6:46PM
		71.10.106.181			99.155.155.225
Row 4		1331261196	source_ip		65.87.165.114
		3/8/2012 6:46PM			71.10.106.181
		76.102.156.138			76.102.156.138

¹ <https://arrow.apache.org/overview/>

Data manipulation

```
imdb = load_dataset("imdb", split="train")  
  
# Filter imdb  
filtered = imdb.filter(lambda row: row['label']!=0)
```

```
Dataset({  
  features: ['text', 'label'],  
  num_rows: 12500  
})
```

¹ <https://huggingface.co/docs/datasets/process#select-and-filter>

Data manipulation

```
# Slicing  
sliced = filtered.select(range(2))  
  
print(sliced)
```

```
Dataset({features: ['text', 'label'], num_rows: 2})
```

```
print(sliced[0]['text'])
```

```
I rented I AM CURIOUS-YELLOW...
```

¹ <https://huggingface.co/docs/datasets/process#select-and-filter>

Benefits of datasets



- ☐ Accessible and shareable
- ☐ Relevant to common ML tasks
- ☐ Efficient processing on large data

Let's practice!
WORKING WITH HUGGING FACE